



Addressing Task Conflicts in Stuttering Detection via MMoE-Based Multi-Task Learning

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Abstract

Multi-Task Learning (MTL) is widely used in automatic stuttering detection to identify stuttering symptoms; however, task conflicts can hinder performance. This paper addresses the task conflict issue in MTL-based stuttering detection and proposes a rule-based MTL strategy and a Multi-Mixture-of-Experts (MMoE) MTL framework to alleviate these conflicts. We analyze the inherent conflicts in stuttering detection tasks and develop a rule-based MTL strategy to mitigate them. Additionally, we introduce an MoE-based adaptive multi-task strategy to optimize task allocation. Experimental results show that our approach outperforms current state-of-the-art methods. In the 2024 SLT Stuttering Speech Challenge, the rule-based MTL strategy achieved a 19.9% increase in average F1 score over the baseline, securing first place. The MMoE-MTL strategy further enhanced task collaboration, improving the average F1 score by 7.55%, demonstrating its effectiveness.

Index Terms: stuttering, MTL, MMoE

1. Introduction

Stuttering is a common speech disorder that affects approximately 70 million people globally, representing about 1% of the total population [1], [2]. For people who stutter (PWS), stuttering not only impacts their social functioning but may also have negative effects on their mental health. With the advancement of communication technologies, these impacts may be further exacerbated. Therefore, the realization of automated stuttering detection holds significant practical value.

Stuttering symptoms typically manifest as the prolongation, pausing, and repetition of syllables, words, or phrases [3]. These symptoms exhibit both commonalities and differences: from a pathological perspective, they may stem from similar neural mechanisms and often co-occur clinically, showing strong correlations. However, from an acoustic signal perspective, they exhibit significant variations in temporal characteristics. Existing research on automated stuttering detection and classification can be broadly categorized into two approaches: one that focuses on the differences between symptoms, treating stuttering detection as a multi-class single-task problem, and another that focuses on the correlations among symptoms, treating stuttering detection as an MTL problem.

Despite the extensive research on stuttering detection, accurately detecting multiple stuttering tasks remains a

challenge due to the conflicting relationship between their correlations and differences. On one hand, most studies focus solely on the differences between stuttering tasks, overlooking the correlations among them. However, these correlation insights are crucial for stuttering detection, especially in data-scarce scenarios. On the other hand, while some studies have considered the correlations between tasks to improve detection performance, they often neglect the potential conflicts among tasks. In MTL for stuttering detection, there are both shared and conflicting components between tasks. Shared components include temporal dependencies, pronunciation abnormalities, and phoneme-level features, while conflicting components may involve time scales and signal characteristics. When sub-tasks may be unrelated or even contradictory, the overall effectiveness of MTL can be compromised.

To address this, we propose a novel stuttering event detection method aimed at balancing the shared and conflicting components within stuttering MTL. The model first investigates the relationships between tasks in stuttering MTL and then designs a rule-based MTL strategy based on these relationships. Additionally, we introduce an MMoE-based automated MTL strategy. Experimental results demonstrate that our approach outperforms existing state-of-the-art techniques in stuttering detection tasks.

2. Related Work

Most studies have approached stuttering detection as either a multi-class classification problem or a binary classification task for various speech disfluencies. Early works, such as Esmaili et al. [4], used Support Vector Machines (SVM) with Perceptual Linear Prediction (PLP) features to detect speech prolongations. Over time, research has shifted towards more complex models [5], [6], [7], [8], [9], [10], [11]. Long Short-Term Memory (LSTM) networks, in particular, have gained attention for their ability to capture long-term dependencies and contextual features in speech. Gupta et al. [6] combined weighted Mel-Frequency Cepstral Coefficients (MFCC) with Bidirectional LSTM (BiLSTM) networks to propose a stuttering classification model. Kourkounakis et al. [7] further improved performance by integrating Residual Neural Networks (ResNet) with BiLSTM. M. Jouaiti et al. [8] introduced a hybrid CNN-LSTM model that combined traditional acoustic features with deep learning for stuttering detection. These studies highlight the success of advanced neural network architectures in stuttering detection. In recent years, models leveraging pre-trained networks have made significant strides. For instance,

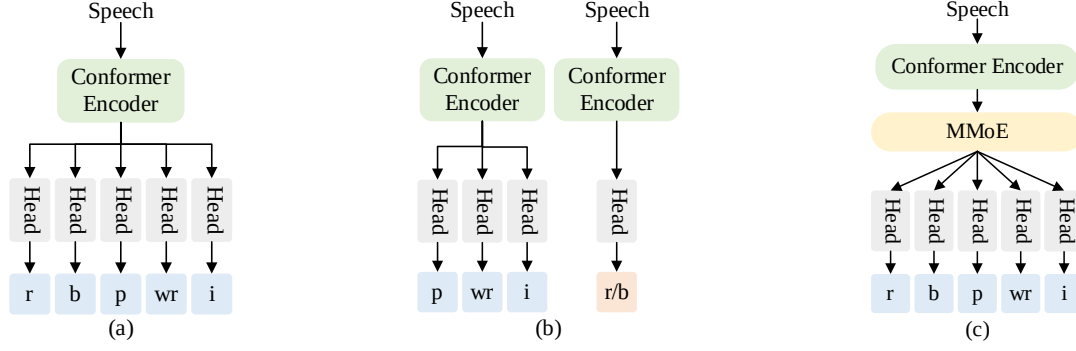


Figure 1: Illustration of the Proposed Multi-Task Learning Strategies. a) Original MTL; b) Rule-based MTL Strategy; c) MTL Based on MMoE. The tasks for stuttering include r, b, p, wr, and i. The head networks used in this study is BiLSTM.

DavFluencyNet [12] utilizes Wav2Vec 2.0 encoder representations and combines convolutional blocks with linear layers for classification. Nie et al. [13] proposed the MMSD-Net model, which incorporates multimodal information and Large Language Models (LLMs) for enhanced detection. Shih et al. [14] focused on word-level stuttering detection, fine-tuning the WavLM model for improved recognition. Single-Task Learning (STL) enables independent learning for each task but overlooks the correlations between stuttering tasks, resulting in a loss of inter-task information. This limits the model’s ability to fully capture stuttering features, affecting detection performance.

Some studies have focused on the correlations between stuttering symptoms and employed Multi-Task Learning (MTL) for modeling [15], [16], [17], [18]. For instance, Lea et al. [15] implemented stuttering detection from podcast data based on an LSTM network. Bayerl et al. [16] constructed the KSoF stuttering dataset and provided strong baseline models. Their subsequent work, combining weighted sums from the pre-trained Wav2Vec 2.0 model with multi-task classification heads, achieved state-of-the-art performance on the KSoF dataset. Furthermore, Sheikh et al. [17] explored the role of MTL and Adversarial Learning (ADV) in robust stuttering feature learning, demonstrating significant improvements in the detection performance of repetitions, blocks, and interjections on the SEP-28k dataset. In general, the adoption of MTL for stuttering detection effectively leverages task-related information to enhance model performance. However, these approaches neglect the potential issue of task conflicts, which may limit further improvements in model performance.

3. Method

3.1. Proposed model

In this study, we use a 12-layer Conformer encoder [19] as the baseline model. The baseline model adopts an MTL framework with five labels, where the head consists of a single-layer BiLSTM network. The specific architecture is illustrated in Figure 1(a). To further optimize model performance, we propose a rule-based MTL approach, in which tasks /p/, /wr/, and /i/ are treated as objectives for MTL, while tasks r and b are trained as independent tasks. The corresponding structure is shown in Figure 1(b). Additionally, we design an MTL model based on a Mixture of Multiple Experts (MMoE) framework, which includes five expert networks: a linear layer, a single-layer BiLSTM, a two-layer BiLSTM, a single-layer

Transformer, and a two-layer Transformer. These expert networks are integrated through a multi-task learning mechanism.

3.2. Multi-Task Learning

During the training process of a stuttering detection system, a key challenge is overfitting and insufficient generalization ability [20]. Overfitting is particularly severe in resource-constrained scenarios, especially when detecting stuttering. To address this issue, multi-task learning (MTL) is an effective regularization method. By introducing an additional auxiliary task that is distinct yet related to the primary task, MTL encourages the model to generalize better. In this paper, we employ an MTL framework with five tasks to solve the stuttering detection problem. Given five tasks $K=(r,b,p,wr,i)$, the model includes a shared network, a Transformer denoted as f , and a head network h^k for each task, where $k \in K$. After the data is processed through the shared network, it is passed to each head network, resulting in the respective outputs y_k for each task. The model for task k can be represented as:

$$y_k = h^k(f(x)) \quad (1)$$

3.3. Multi-gate Mixture-of-Experts

Although the stuttering tasks share information such as speech prosody, there are still significant differences between the tasks, such as inconsistencies in time scales. To maximize the retention of shared information while addressing the differences between tasks, we adopt a Multi-Mixture-of-Experts (MMoE) model [16]. The core idea is to replace the shared bottom network in equation (1) with the original MMoE layer. Additionally, a separate gating network is added for each task. More precisely, the output of the task is given by:

$$y_k = h^k(f^k(x)), \quad (2)$$

$$\text{where } f^k(x) = \sum_{i=1}^n g^k(x)_i f_i(x) \quad (3)$$

where n is the number of experts and g represents the gating network.

3.4. Loss function

BCEWithLogitsLoss (Binary Cross-Entropy with Logits Loss) is a loss function specifically designed for binary classification tasks. For a given sample x_i with the true label $y_i \in \{0,1\}$ the model’s goal is to predict the probability $p_i \in [0,1]$ that the

sample belongs to the positive class. The mathematical expression for BCEWithLogitsLoss is as follows:

$$\text{loss} = \frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\sigma(p_i)) + (1 - y_i) \cdot \log(1 - \sigma(p_i))] \quad (4)$$

where N is the number of samples, $\sigma(x) = 1 / (1 + e^{-x})$ is sigmoid function, and $\log$ is natural logarithm.

4. Experimental Setup

4.1. Dataset

In this paper, the proposed method was evaluated using the AS-70 dataset[21]. The dataset comprises recordings of adults who stutter (AWS) and native Mandarin speakers. A total of 70 native Mandarin-speaking AWS participated in the recording sessions, including 24 females. Each participant underwent an hour-long recording session, consisting of two parts: dialogues and reading speech commands. The annotation guidelines specify five types of stuttering, including:

/wr: Word/phrase repetition, used to mark the entire repetition of a character or phrase.

/b: Block, indicating a pause due to breath or stuttering.

/p: Prolongation, referring to elongated phonemes.

/r: Sound repetition, indicating repeated phonemes that do not constitute a complete character.

/i: Interjection, representing filler characters caused by stuttering, such as 'uh', 'um', or 'er'.

4.2. Experimental Setup

The model is built using the Wenet[22] library for feature extraction and audio processing. We used the Adam optimizer with a learning rate of 0.0001, combined with weighted entropy, focal loss, and an early stopping strategy with 10 patience epochs for model training. In this study, we followed the dataset split method used in the SLT2024 competition to divide the training, validation, and test sets, as shown in Table 1.

Table 1: Division and Proportion of Dataset.

Dataset	number of people	number of files
Train	43(61.42%)	26659(64.09%)
Dev	7(10%)	4294(10.23%)
test	20(28.57%)	11000(26.21%)

4.3. Evaluation

To evaluate the model's performance, we used commonly employed metrics in speech detection: Micro-F1, as shown in formula (5). Additionally, to assess the overall performance of the model across all stuttering tasks, the average F1 score of all tasks, denoted as $F1_{final}$, was used, as shown in formula (6).

$$F1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (5)$$

$$F1_{final} = \text{mean}(F1_p, F1_b, F1_r, F1_{wr}, F1_i) \quad (6)$$

5. Results Analysis

This section provides a detailed analysis of the experimental results for the proposed stuttering event detection model, focusing on the following aspects: first, an ablation study to establish a stable and reliable baseline; second, an analysis of

task conflicts in MTL and a comparison of performance differences among various MTL strategies; and finally, a comprehensive comparison between the proposed method and existing approaches.

5.1. Baseline

To establish a reliable benchmark, this study optimized the baseline model in three key areas: 1) incorporating a pre-trained ASR model, 2) testing different shared bottom layer configurations, and 3) experimenting with various head-layer structures. Table 2 presents the impact of different shared network and head-layer configurations on stuttering task performance. The shared configurations involve varying the number of Conformer layers (3, 6, 12, and 15) and using a pre-trained ASR model, while the head-layer configurations explore different BiLSTM layer counts (1, 2, and 3). The results show: 1) the ASR pre-trained model significantly outperforms the non-pre-trained version, 2) the 12-layer Conformer model performs best than other layers, and 3) a single-layer BiLSTM surpasses multi-layer BiLSTM configurations. Based on these findings, the study adopts the Conformer (12-layer) - BiLSTM (1-layer) - MTL architecture as the baseline for further experiments.

Table 2: $F1_{final}$ Score Comparison Across Shared Network Configurations and Head-Layer Settings for Stuttering Task Performance.

Shared layer		Head layer	$F1_{final}$
Layer num	Pretrained		
Conformer 3	N	N	51.27
Conformer 3	Y	N	58.13
Conformer 6	Y	N	62.99
Conformer 9	Y	N	62.69
Conformer 12	Y	N	63.29
Conformer 15	Y	N	62.94
Conformer 12	Y	BiLSTM 1	64.64
Conformer 12	Y	BiLSTM 2	64.31
Conformer 12	Y	BiLSTM3	62.78

5.2. Multi-Task Learning Strategies

5.2.1. Task Conflict Similarity Analysis

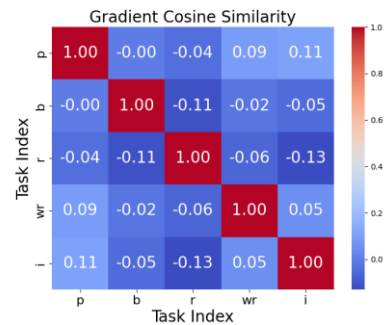


Figure 2: Heatmap of the inter-task gradient cosine similarity using the Conformer-BiLSTM-MTL framework.

In the MTL framework, low correlation between tasks can lead to interference among shared parameters, thereby affecting model performance. Particularly in the stuttering detection task, there are significant differences in the time scales of different tasks. To verify whether this inconsistency triggers multi-task

conflicts, this study quantifies the degree of task conflict by calculating the gradient descent cosine similarity. The experimental results are shown in Figure 2. Analysis indicates that there are indeed significant gradient conflicts between tasks: specifically, the /b and /r tasks exhibit strong conflicts with all other tasks, while the /p, /i, and /wr tasks show relatively smaller conflicts. This finding provides important evidence for the subsequent design of targeted multi-task optimization strategies.

5.2.2. Multi-task strategy comparison

To mitigate the negative impact of task conflicts on model performance in MTL, this study proposes two innovative solutions: a rule-based MTL strategy and an MTL method based on Multi-Mixture-of-Experts (MMoE). Table 3 compares the performance of four training strategies: (1) single-task independent training (No-MTL); (2) original multi-task learning (Origin-MTL); (3) rule-based MTL strategy (Rule-MTL); and (4) MMoE-based MTL method (MMoE-MTL). The rule-based MTL strategy employs a grouped training scheme, where the /p, /wr, and /i tasks form a three-task MTL, while the /b and /r tasks are trained as independent models.

Table 3: Performance Comparison in F1 score of Different Multi-Task Strategies Using a 12-Layer Conformer Base with a BiLSTM Head Layer.

MTL strategy	Task label				
	/p	/b	/r	/wr	/i
No-MTL	66.55	44.07	59.78	70.75	29.77
Origin-MTL	67.90	38.17	58.78	73.75	84.64
Rule-MTL	70.89	44.07	59.78	74.79	85.15
MMoE-MTL	68.68	70.81	69.80	71.09	79.58

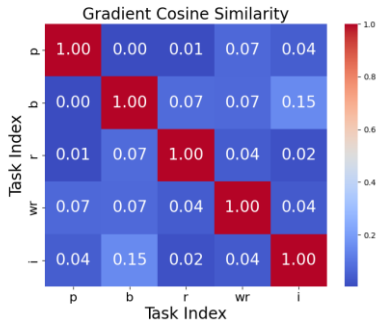


Figure 3: Heatmap of the inter-task gradient cosine similarity using the Conformer-MMoE-BiLSTM-MTL framework.

The experimental results indicate that: First, comparing No-MTL and Origin-MTL, the original MTL improves the performance of the /p, /i, and /wr tasks, but significantly reduces the performance of the /b and /r tasks, confirming the presence of task conflicts. Second, comparing Origin-MTL with Rule-MTL shows that grouping /p, /wr, and /i into a three-task MTL yields better performance than the five-task MTL, validating that these three tasks are more highly correlated. Finally, a comparison of the performance between Rule-MTL and MMoE-MTL reveals that the MMoE method achieves significant improvements in the /b, /r, and /wr tasks, with particularly notable performance gains observed in the /b task. These results suggest that MMoE not only effectively mitigates conflicts arising from other tasks that hinder the /b task, but also

provides the shared information necessary for the /b task. Furthermore, the gradient descent similarity analysis further corroborates this conclusion. As shown in Figure 3, compared to Origin-MTL in Figure 2, MMoE-MTL exhibits lower gradient conflicts, providing strong evidence for the effectiveness of the MMoE method.

5.3. State-of-the-art results

To comprehensively evaluate the performance advantages of the proposed method, this study conducted a systematic comparison with existing state-of-the-art methods. The baseline [18] refers to the results of a three-layer Conformer applied to multi-task learning. T018[18] uses external normal data to simulate stuttering data and incorporates the Zipformer architecture. T031[18] proposes a fine-grained contrastive learning (FGCL) framework and introduces a novel mining algorithm to identify simple and easily confused frames in audio data. As shown in Table 4, the results demonstrate that the proposed method significantly outperforms existing approaches in F1 scores for core tasks such as /b/, /p/, /r/, and /i/. Furthermore, the $F1_{final}$ score shows an 11.27% improvement over the best-performing model, T018[18]. These findings strongly validate the effectiveness and superiority of the proposed stuttering event detection method.

Table 4: Performance Comparison in F1 score of the Proposed Method with State-of-the-Art Approaches.

Model	Task label				
	/p	/b	/r	/wr	/i
Baseline[18]	65.12	24.3	41.86	61.85	74.87
StutterNet[9]	59.93	45.05	54.46	50.83	63.54
ConvLSTM[15]	32.66	25.24	31.24	64.51	50.54
T018[18]	67.61	31.58	59.25	81.11	83.94
T031[18]	65.89	24.78	50.15	64.85	77.06
Rule-MTL	70.89	44.07	59.78	74.79	85.15
MMoE-MTL	68.68	70.81	69.80	71.09	79.58

6. Conclusions

This study systematically explores the task conflict issue in automatic stuttering detection within the multi-task learning (MTL) framework and proposes rule-based and Multi-Mixture-of-Experts (MMoE)-based MTL frameworks to effectively alleviate conflicts between tasks. By using the Conformer network to extract shared acoustic features and combining the rule-based MTL strategy with the MMoE-based adaptive multi-task strategy, we successfully enhanced the performance of stuttering detection. Experimental results show that the proposed method achieved significant outcomes in the 2024 SLT Stuttering Speech Challenge, with the rule-based MTL strategy improving the average F1 score on the AS-70 dataset by 19.9% compared to the baseline model, and the MMoE-based strategy further increasing the F1 score by 7.55%. These results not only validate the effectiveness of the proposed method but also provide new insights for the future application of multi-task learning in speech processing.

Future work will focus on analyzing the deeper relationships between tasks in stuttering detection to improve interpretability from a theoretical perspective. Additionally, we plan to further optimize model performance by integrating acoustic signals with semantic information, and explore the model's application and scalability on other stuttering datasets.

7. Acknowledgements

This research is supported by National Natural Science Foundation of China (U23B2018), National Natural Science Foundation of China (NSFC 62271477), Shenzhen Science and Technology Program (JCYJ20220818101411025), Shenzhen Science and Technology Program (JCYJ20220818102800001), Shenzhen Science and Technology Program (JCYJ20220818101217037), Shenzhen Peacock Team Project (KQTD20200820113106007), Shenzhen Medical Research Fund (C2401001), Shenzhen University-Huaqiang Project.

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